Sr. No._____ Date:_____

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Section: 23BCS_ Krg. 13
Subject: Competitive ~~pro~~ coding

Q Given three integers n, a & b, return nth magical no. Since the array maybe very large return 10^9+7 mod.

A positive integer is magical if the number is divisible by either a or b.

n = 1, a = 2, b = 3
ans = 2

n = 4, a = 2, b = 3
ans = 6

⇒ Brute force, Time complexity, O(n * mo.(a,b))

1) Start
2) Initialize and set
count = 0 & num = 1
3) while (count < n) {
if ((num % a == 0) || (num % b == 0)) {
count ++
}
num ++
}
4) Initialize & set ans = num - 1
5) Return ans % 1e9 + 7
6 Exit

Evergreen

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⇒ Optimal approach, Time complexity: ~~O(log n)~~
$O(n * \min(a,b))$

1) Start
2) calculate gcd(a,b)
3) Then calculate the lcm
using $lcm(a,b) = \frac{a*b}{gcd(a,b)}$

4) Now, set $l = 1$, $r = n * \min(a,b)$
5) while $(l \le r)$ {
   a) $mid = (l+r)/2$
   b) $count = \frac{mid}{a} + \frac{mid}{b} - \frac{mid}{lcm}$

   c) if $count \ge k$
      $ans = mid$
      $r = mid - 1$

   d) else
      $l = mid + 1$

6) Return $ans \% (10^9 + 7)$
7) Exit